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Installing SWI-Prolog on Windows

A simple guide for knowledge representation tasks

1. Download

1. Open the stable download page: **https://www.swi-prolog.org/download/stable**
2. Under Windows, click **Windows 64-bit installer** (file name like swipl-x.y.z-1.x64.exe).

Note: the daily builds page (`swi-prolog.org/download/daily/bin/`) also works, but those are development snapshots and can be unstable. For coursework, use the stable release.

2. Install

1. Double-click the downloaded .exe file.
2. Click Yes if Windows asks for permission.
3. On the options screen, tick **Add swipl to the system PATH**.
4. Keep the option to associate **.pl** files with SWI-Prolog.
5. Click Next until the installation finishes, then click Finish.

3. Check that it works

1. Open the Start menu and search for Command Prompt.
2. Type this command and press Enter:

```
swipl --version
```

A version number should appear. If Windows says the command is not found, close the window and open a new one, or start the SWI-Prolog app from the Start menu.

4. Write a knowledge base

Open Notepad and paste the text below. Save it as family.pl (choose "All files" as the type so it does not become family.pl.txt).

```
% Facts
parent(ali, sara).
parent(ali, omar).
parent(sara, noor).
male(ali).
male(omar).
female(sara).
female(noor).

% Rules
father(X, Y) :- parent(X, Y), male(X).
```

```
grandparent(X, Z) :- parent(X, Y), parent(Y, Z).
sibling(X, Y) :- parent(P, X), parent(P, Y), X \= Y.
ancestor(X, Y) :- parent(X, Y).
ancestor(X, Y) :- parent(X, Z), ancestor(Z, Y).
```

Every fact and rule must end with a period.

5. Load the file and ask questions

1. Open Command Prompt and go to the folder where you saved the file:

```
cd C:\Users\YourName\Documents
```

2. Start Prolog and load the file:

```
swipl family.pl
```

3. Type queries at the `?-` prompt:

```
?- grandparent(ali, noor).
?- sibling(sara, X).
?- ancestor(ali, Who).
```

Press; for the next answer, and **Enter** to stop. You can also double-click family.pl to open it directly in SWI-Prolog.

6. Useful commands

Command	Purpose
`make.`	Reload the file after editing
`listing(parent).`	Show stored facts and rules
~~`assert(parent(omar, lina)).`~~	~~Add a fact~~
~~`retract(parent(omar, lina)).`~~	~~Remove a fact~~
`halt.`	Exit Prolog

7. Common problems

- **Unknown procedure:** a typo in a predicate name, or the file is not loaded.
- **Syntax error:** usually a missing period. Variables start with a capital letter (X), constants with lowercase (ali).
- **Singleton variable warning:** a variable is used only once. Replace it with _ if you do not need it.
- **File saved as family.pl.txt:** rename it to family.pl.